

Umelá inteligencia Zadanie 7

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Cieľ úlohy

Vytvorte a natrénujte dve neurónové siete pre rozpoznávanie rukou písaných číslíc: viacvrstvovú perceptrónovú sieť (MLP) a konvolučnú neurónovú sieť (CNN). Cieľom je porovnať klasický ANN prístup a deep learning prístup na rovnakom datasete.

Pre MLP musí mať najlepšia z porovnávaných štruktúr na testovacích dátach priemernú úspešnosť aspoň 97%, pričom priemer sa určí z 5 nezávislých spustení trénovania odznova. Pre CNN musí mať najlepšia z porovnávaných štruktúr na testovacích dátach priemernú úspešnosť aspoň 99%, opäť ako priemer z 5 nezávislých spustení trénovania odznova.

DEVICE: mps

train samples: 49999

val samples : 10001

test samples : 10000

MLP a hlavné hyperparametre

Model	Skryté Vrstvy	Neuróny	Epochy	Learning Rate	Batch
MLP1	1	128	15	0.001	128
MLP2	2	256, 128	15	0.001	128

MLP1

Beh	Train loss	Test loss	Train acc [%]	Test acc [%]
1	0.031	0.079	99.2	97.6
2	0.024	0.075	99.4	97.8
3	0.025	0.081	99.4	97.6
4	0.031	0.082	99.2	97.6
5	0.044	0.081	99.8	97.6

MLP2

Beh	Train loss	Test loss	Train acc [%]	Test acc [%]
1	0.025	0.076	99.2	98.0
2	0.022	0.072	99.3	97.9
3	0.041	0.069	99.7	97.9
4	0.030	0.082	99.0	97.5
5	0.021	0.087	99.4	97.8

MLP1 vs. MLP2

Model	Min. test [%]	Max. test [%]	Avg test [%]	Avg test loss
MLP1	97.6%	97.8%	97.6%	0.080
MLP2	97.5%	98.0%	97.8%	0.077

CNN a hlavné hyperparametre

Model	Conv Vrstvy	Filtre	FC vrstvy	Epochy	Batch
CNN1	2	32, 64	128	15	128
CNN2	3	32, 64, 128	128	15	128
CNN3	3	32, 64, 128	256	15	128

CNN1

Beh	Train loss	Test loss	Train acc [%]	Test acc [%]
1	0.014	0.026	99.5	99.1
2	0.009	0.028	99.7	99.2
3	0.023	0.033	99.3	98.8
4	0.029	0.032	99.1	98.9
5	0.015	0.029	99.5	99.0

CNN2

Beh	Train loss	Test loss	Train acc [%]	Test acc [%]
1	0.028	0.032	99.1	99.0
2	0.015	0.025	99.5	99.2
3	0.013	0.032	99.5	99.1
4	0.017	0.026	99.5	99.1
5	0.017	0.027	99.4	99.1

CNN3

Beh	Train loss	Test loss	Train acc [%]	Test acc [%]
1	0.027	0.030	99.1	99.0
2	0.015	0.028	99.5	99.2
3	0.012	0.030	99.6	99.1
4	0.024	0.023	99.2	99.2
5	0.018	0.025	99.4	99.2

CNN1 vs. CNN2 vs. CNN3

Model	Min. test [%]	Max. test [%]	Avg test [%]	Avg test loss
CNN1	98.8	99.2	99.0	0.030
CNN2	99.0	99.2	99.1	0.028
CNN3	99.0	99.2	99.1	0.027

Dropout

Model	Dropout	Počet behov	Priemer epochy začiatku pretrénovania	Priemer val loss
D1	0.1	5	9.2	0.075
D2	0.3	5	11.0	0.073
D3	0.5	5	13.2	0.079

Porovnanie modelov s rôznym nastavením dropout-u.

Dropout	Počet behov	Avg train acc [%]	Avg test acc [%]	Avg test loss
0.1	5	99.0	98.0	0.069
0.3	5	98.3	98.0	0.073
0.5	5	97.2	97.8	0.074

Súhrnné porovnanie priemerných výsledkov pre rôzne nastavenia dropout-u.

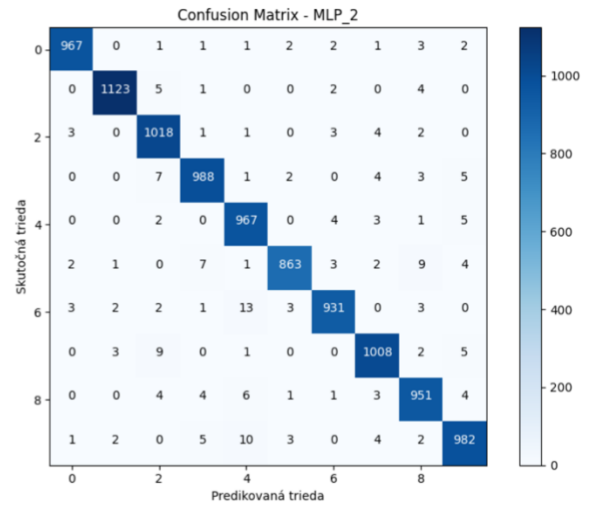
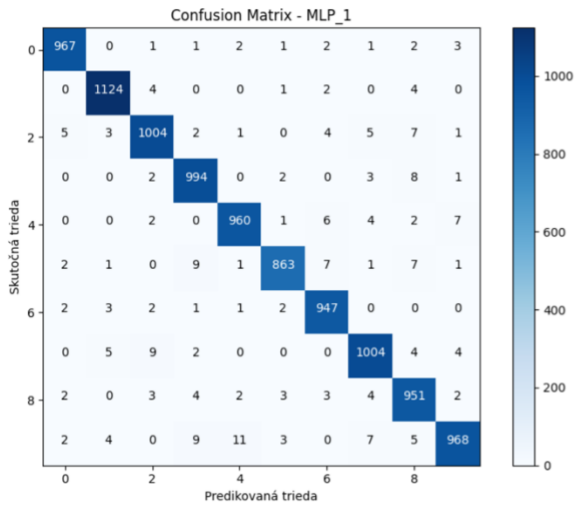
Porovnanie

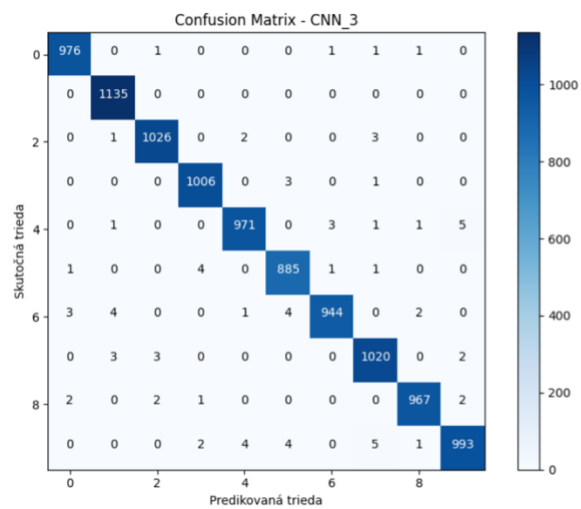
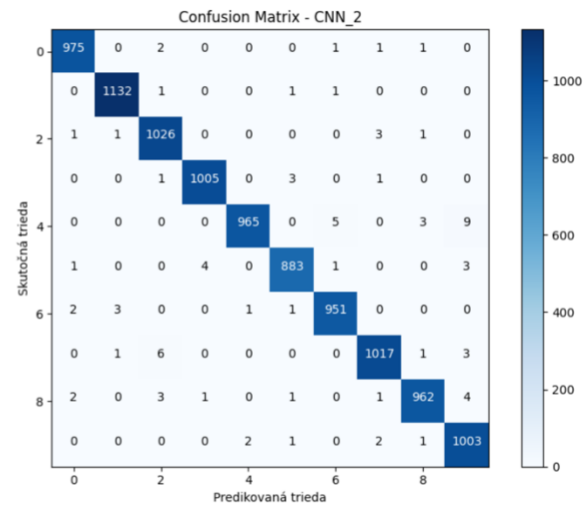
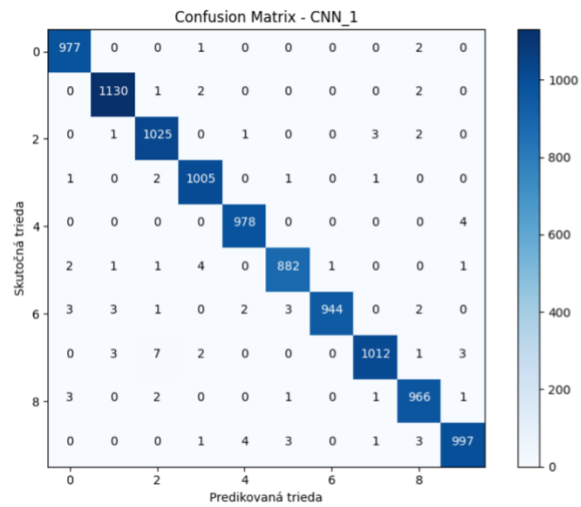
Prístup	Počet modelov	Avg test acc [%]	Avg test loss
MLP	2	97.7	0.0785
CNN	3	99.06	0.0283

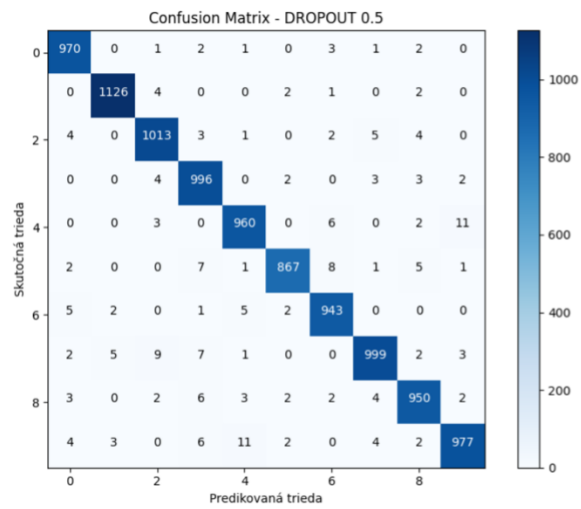
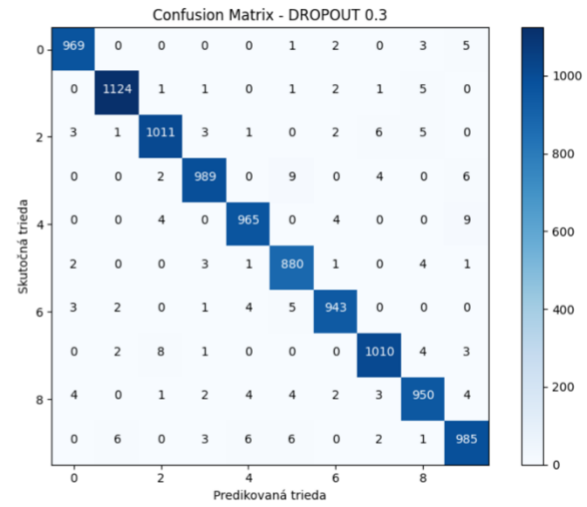
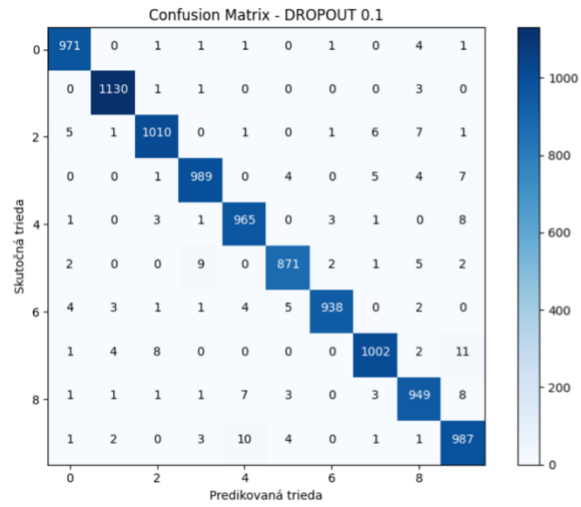
Súhrnné porovnanie priemerných výsledkov MLP a CNN.

Model	CPU čas [s]	GPU čas [s]	Zrýchlenie
MLP2	1.7	2.4	0.7x
CNN2	23.3	5.1	4.5x
Priemer (Avg)	12,5	3,75	2,6x

Súhrnné porovnanie času trénovania na CPU a GPU.







MLP1:



MLP2:



CNN1:



CNN2:



CNN3:



D1:



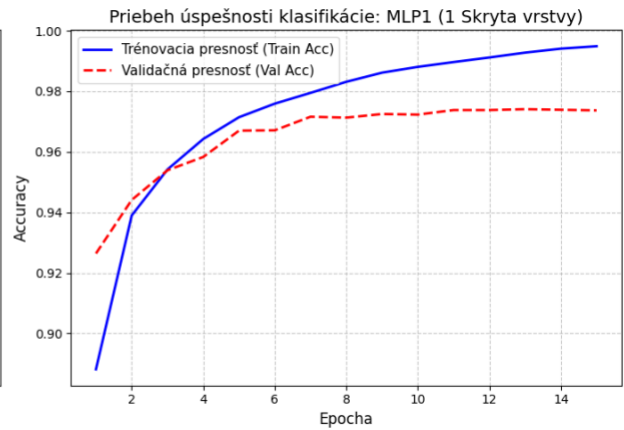
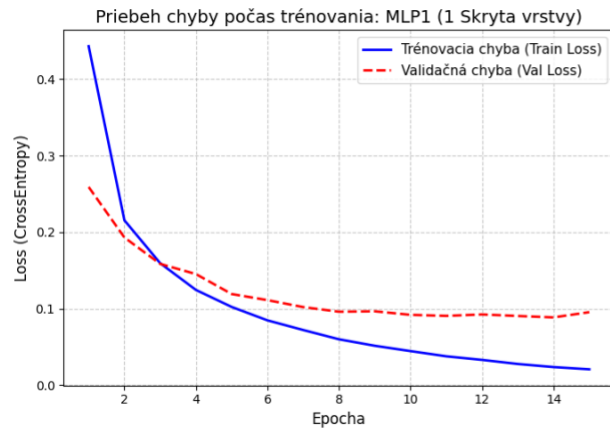
D2:



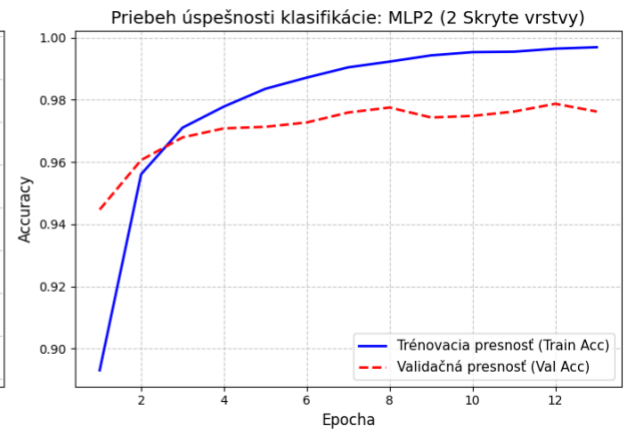
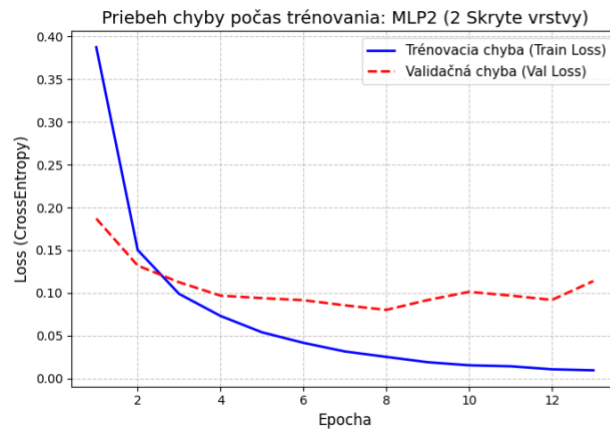
D3:



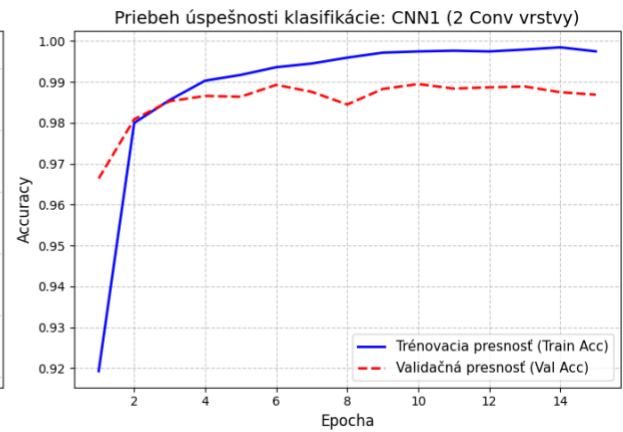
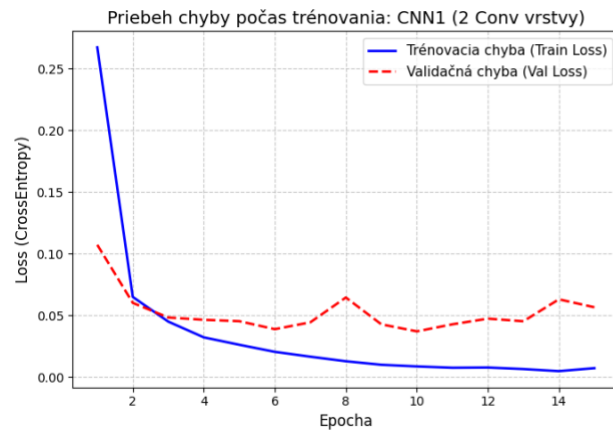
MLP1



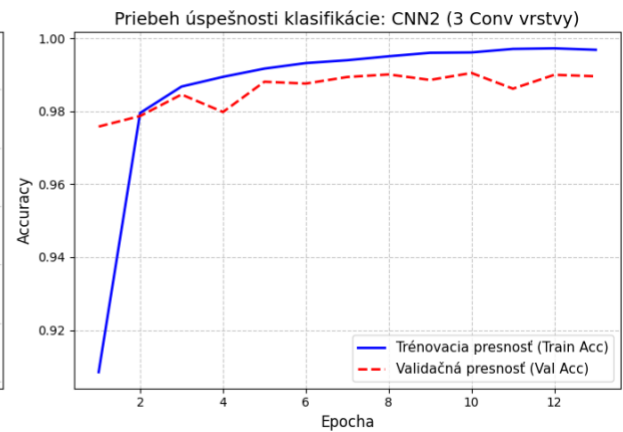
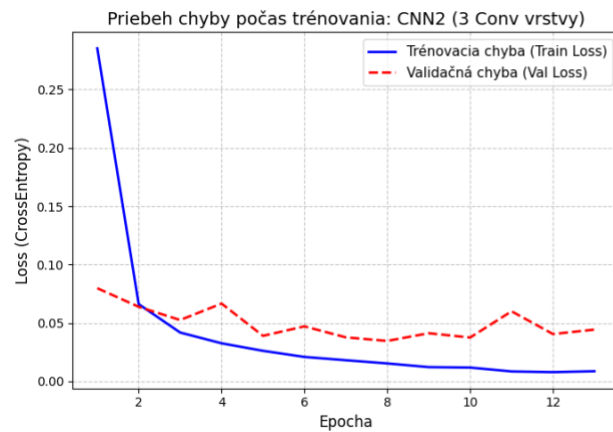
MLP2



CNN1



CNN2



CNN3

